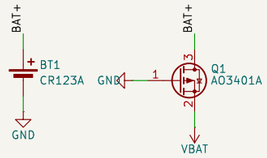
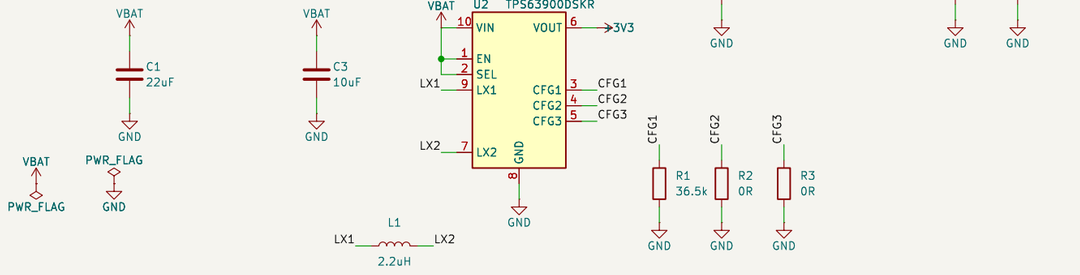
1 Battery CR123A (in the enclosure, wired to BAT+/BAT−), reverse polarity protection

Q1: drain to the cell, source to the load, gate to GND (a reversed cell turns it off)



2 TPS63900 buck-boost, 3.3 V out, 75 nA quiescent

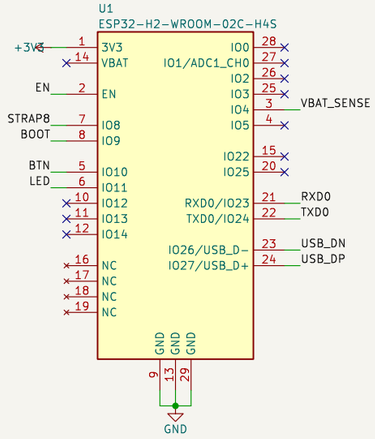


CFG1 36.5k + CFG2 0R: VO(2) = 3.3 V, no input current limit; CFG3 0R: VO(1) = 1.8 V, unused: EN, SEL to VIN

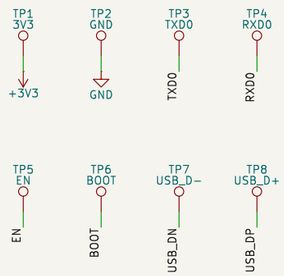


3 ESP32-H2-WROOM-02C

Pin 14 VBAT: leave open, tied to 3V3 inside the module

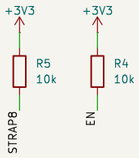


4 Test pads (bottom side, pogo pins); current: ammeter in the BAT+ wire

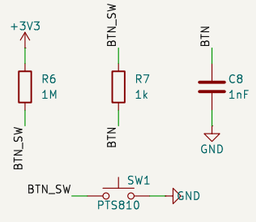


5 EN pull-up, strapping

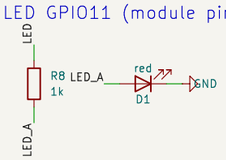
No EN capacitor (the module has its own 10k/1uF RC)



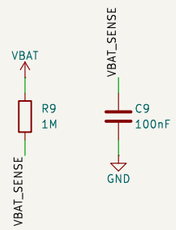
6 Button GPIO10 (LP IO, EXT1 wake-up)



7 LED GPIO11 (module pin 6)



8 Battery sense GPIO4 (ADC1_CH3, module pin 3), no divider



Generated by ../scripts/gen_sch.py from design.py
CR123A -> Q1 (reverse polarity) -> TPS63900 3.3 V -> ESP32-H2-WROOM-02C
Haxeko

Sheet: /		
File: matter-button-module-cr123a-mini.kicad_sch		
Title: ESP32-H2 Matter over Thread button, CR123A, 34.5 mm board		
Size: A3	Date: 2026-11-12	**Rev: 1.0**
KiCad E.D.A. 9.0.9		Id: 1/1